

1. **Week-7:** Write a script to create a topic with specific partition and replication factor settings.

week-7/

└─ kafka-producer-demo/

└─ docker-compose.yml

└─ create_topic.sh

Expected output:

```
kafka-producer-demo/  
└─ create_topic.sh  
└─ docker-compose.yml
```

docker-compose.yml

version: '3.8'

services:

zookeeper:

image: confluentinc/cp-zookeeper:7.4.0

environment:

ZOOKEEPER_CLIENT_PORT: 2181

ZOOKEEPER_TICK_TIME: 2000

ports:

- 2181:2181

kafka:

image: confluentinc/cp-kafka:7.4.0

depends_on:

- zookeeper

ports:

- 9092:9092

- 9093:9093

environment:

KAFKA_BROKER_ID: 1

KAFKA_ZOOKEEPER_CONNECT: 'zookeeper:2181'

KAFKA_LISTENER_SECURITY_PROTOCOL_MAP:
PLAINTEXT:PLAINTEXT,PLAINTEXT_HOST:PLAINTEXT

KAFKA_LISTENERS: PLAINTEXT://0.0.0.0:9092,PLAINTEXT_HOST://0.0.0.0:9093

KAFKA_ADVERTISED_LISTENERS:
PLAINTEXT://kafka:9092,PLAINTEXT_HOST://localhost:9093

KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR: 1

STEP 4 :- docker-compose up -d

nano create_topic.sh

```
#!/bin/bash
```

```
TOPIC_NAME="my-topic"
```

```
PARTITIONS=3
```

```
REPLICATION_FACTOR=1
```

```
CONTAINER_NAME="kafka-producer-demo-kafka-1" # container name from docker ps
```

```
echo "Creating Kafka topic: $TOPIC_NAME ..."
```

```
docker exec $CONTAINER_NAME kafka-topics --create \
```

```
--topic $TOPIC_NAME \
```

```
--bootstrap-server localhost:9092 \  
--partitions $PARTITIONS \  
--replication-factor $REPLICATION_FACTOR
```

echo "Verifying topic creation..."

```
docker exec $CONTAINER_NAME kafka-topics --describe \  
--topic $TOPIC_NAME \  
--bootstrap-server localhost:9092
```

Run Terminal:-

```
$ chmod +x create_topic.sh
```

```
chmod +x create_topic.sh
```

```
./create_topic.sh
```

Output:-

Created topic my-topic.

Verifying topic creation...

Topic: my-topic TopicId: HXx3_QJ6QmSbMa3RwCxL2A PartitionCount: 3 ReplicationFactor:

1 Configs:

Topic: my-topic Partition: 0 Leader: 1 Replicas: 1 Isr: 1

Topic: my-topic Partition: 1 Leader: 1 Replicas: 1 Isr: 1

Topic: my-topic Partition: 2 Leader: 1 Replicas: 1 Isr: 1